

INTERNSHIP

by Pushpal Das

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ABSTRACT

As electronic devices become more complex and ubiquitous in our daily lives, the reliability and performance of printed circuit boards (PCBs) and printed wiring assemblies (PWAs) become increasingly important. Failure analysis of PCBs and PWAs is a critical aspect of ensuring their reliability, and it involves the systematic investigation of the causes of failures that occur during design, manufacturing, or operation. There are various failure mechanisms that can occur in PCBs and PWAs. Component failures are common, and they can be caused by factors such as temperature, humidity, and mechanical stress. Solder joint failures are another common failure mechanism, and they can result from poor soldering quality, thermal cycling, or mechanical stress. Trace and via failures can also occur due to issues such as manufacturing defects, electrostatic discharge, and corrosion. To conduct a failure analysis, various tools and techniques are used, depending on the nature and extent of the failure. Visual inspection is often the first step, and it involves examining the failed component or assembly for any visible signs of damage or defects. X-ray imaging is another useful technique for identifying defects in PCBs and PWAs, particularly for identifying issues with internal components, such as BGA packages. Micro sectioning is also commonly used to investigate the microstructure of solder joints or other components, and it can help identify the root cause of a failure. Once the root cause of a failure has been identified, steps can be taken to mitigate the risk of similar failures occurring in the future. This may involve changes to the design, materials, or manufacturing processes, or it may involve changes to the testing and inspection protocols to ensure that potential issues are caught before the device is released into the field. In conclusion, failure analysis of PCBs and PWAs is critical for ensuring the reliability and performance of electronic devices. By understanding the various failure mechanisms and using appropriate tools and techniques to investigate them, engineers can identify the root cause of a failure and take steps to prevent similar issues from occurring.

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To
Pushpal Das
SRM Institute of Science and Technology
Chennai, India

Date: 16 November 2022

Subject: Internship Offer Letter

Dear Pushpal,

Based on our evaluation during the internship recruitment campus, we are pleased to inform you that you have been selected as an Intern by SLB.

We are pleased to offer you internship with Schlumberger India Technology Centre Pvt. Ltd effective 02 January 2022 on successfully passing your pre-employment medical.

The duration of the internship will be 6 months and you will be eligible to receive below compensation components during the internship, subject to fulfillment of the associated tasks and formalities.

One time Travel Expenses: INR 16,000
One time accommodation expenses: INR 35,000
Monthly Stipend: INR 25,000
Monthly Housing Allowance: INR 15,000
Monthly Commuting Allowance: INR 5,000

The offer is subject to satisfactory completion of:

- A medical examination by a doctor to present fitness certificate, which should be conducted prior to the commencement of the Internship.
- Signing of the internship agreement at the time of commencement of internship.

Please sign, date and return a scanned copy of this letter within one week of receipt to vpandya@slb.com to acknowledge your acceptance of this internship offer.

Company Identification Number (CIN): U29299PN2008PTC131255 **SLB-Private**



By accepting this offer letter you confirm that you will comply with applicable SLB policies during the course of your internship. Please note that this offer letter is for the requested internship only and does not create any employment or guarantee future employment with SLB.

We would like to take this opportunity to welcome you to SLB and wish you success in your future career.

Yours Sincerely,

A handwritten signature in black ink, appearing to read 'Ruchir'.

Ruchir Srivastava
STSI HR Manager

I, _____, accept the internship offer and terms and conditions mentioned above.

Signature _____ Date _____

SLB-Private

X

SLB-Private

CHAPTER 1

INTRODUCTION

1.1 Company Background

Schlumberger is a multinational oilfield services company headquartered in Paris, France. The company was founded in 1926 by Conrad and Marcel Schlumberger, brothers who invented a new method for measuring the electrical resistivity of rocks. The company operates in more than 120 countries, employing over 80,000 people. Schlumberger's services span the entire lifecycle of an oil or gas field, from exploration and drilling to production and maintenance. The company provides advanced technology and expertise in areas such as reservoir characterization, drilling, well construction, and production optimization. Schlumberger also offers software and data management solutions to help its clients make better decisions and increase efficiency. Schlumberger is committed to sustainability and reducing its environmental impact. Schlumberger is also involved in several social and community initiatives, such as supporting STEM education programs and providing disaster relief assistance. Overall, Schlumberger has a long history of innovation and leadership in the oilfield services industry. With a global presence and a strong focus on research and development, the company is well-positioned to continue providing essential services to the oil and gas industry while adapting to the changing needs of the energy transition.

Digital and Online Platforms

Schlumberger has a range of digital and online platforms that support its oilfield services and provide valuable information to its clients. Some of these platforms include:

- DELFI: This is Schlumberger's integrated cloud-based platform that provides access to data and analytics for oil and gas operations. DELFI enables data-driven decision-

making by bringing together data from multiple sources and providing tools for analysis and collaboration.

- Petrel: This is Schlumberger's software platform for reservoir modeling and characterization. Petrel enables geoscientists and engineers to build accurate 3D models of reservoirs and simulate production scenarios.
- INTERSECT: This is Schlumberger's high-resolution reservoir simulator that enables accurate modeling of complex reservoirs. INTERSECT uses advanced algorithms and parallel computing to deliver fast and accurate simulation results.
- GeoFrame: This is Schlumberger's platform for geoscience and reservoir engineering workflows. GeoFrame provides a range of tools for data integration, interpretation, and modeling.

1.2 Company's Operations

Operations span the entire lifecycle of an oil or gas field, from exploration and drilling to production and maintenance. Some of the key areas of Schlumberger's operations include:

Reservoir Characterization: Schlumberger provides advanced technology and expertise for characterizing and understanding oil and gas reservoirs. This includes services such as seismic imaging, well testing, and formation evaluation.

Drilling: The company's drilling technologies are designed to improve drilling efficiency and reduce costs.

Well Construction: Schlumberger provides a range of well construction services, including cementing, completions, and well intervention. These services help ensure that wells are constructed safely and efficiently.

Production: Schlumberger offers technology and expertise for optimizing oil and gas production. This includes services such as well performance analysis, artificial lift systems, and production logging.

Integrated Project Management: Schlumberger provides project management services for complex oilfield projects.

1.3 Future Outlook

Schlumberger has a strong outlook for the future, as the company continues to innovate and adapt to the changing needs of the energy industry. Some key factors that are expected to shape the future of Schlumberger include:

Energy Transition: Schlumberger recognizes the importance of the energy transition and is committed to reducing its carbon footprint and developing new technologies that support the transition to a lower carbon economy. The company is investing in technologies such as carbon capture and storage, geothermal energy, and hydrogen production.

Digitalization: Schlumberger is at the forefront of digitalization ⁵ in the oil and gas industry, and the company is expected to continue to innovate in this area. The company's digital platforms, such as DELFI and Petrel, are expected to become even more important in the future as data and analytics become increasingly critical to oil and gas operations.

Research and Development: Schlumberger has continued to invest in new technologies and innovations that can improve the efficiency and productivity of oil and gas operations.

Industry Consolidation: The oilfield services industry is highly competitive, and Schlumberger is expected to continue to consolidate its position through acquisitions and partnerships with other companies.

1.5 Awards and Recognition

Some of the most notable awards and recognitions received by Schlumberger include:

Forbes Global 2000 List: Schlumberger has been included in the Forbes Global 2000 list of the world's largest public companies multiple times.

Energy Intelligence Top 100 List: Schlumberger has been ranked in the top 10 of the Energy Intelligence Top 100 list of the world's largest oil and gas companies.

Dow Jones Sustainability Index: Schlumberger has been included in the Dow Jones Sustainability Index, which recognizes companies for their sustainability performance and commitment.

Fortune World's Most Admired Companies List: Schlumberger has been included in Fortune's World's Most Admired Companies list multiple times, recognizing the company's reputation and performance.

Global Energy Awards: Schlumberger has won multiple Global Energy Awards, including the Energy Company of the Year and the Oilfield Services Company of the Year awards.

CHAPTER 2

TECHNOLOGIES AND FRAMEWORKS

2.1 Reservoir performance

Reservoir performance is a critical aspect of industry, as it determines the efficiency and productivity of oil and gas operations. Reservoir performance refers to the ability of the reservoir to produce hydrocarbons, which depends on various factors such as the reservoir rock properties, fluid characteristics, and production techniques.

To optimize reservoir performance, the oil and gas industry relies on advanced technologies and frameworks that enable better understanding and management of reservoirs. Some key technologies and frameworks used in reservoir performance management include:

Reservoir Modeling: Reservoir modeling involves creating a computerized model of the reservoir based on geological and engineering data.

Production Logging: Production logging involves measuring various parameters such as flow rate, pressure, and temperature in the wellbore to determine the production characteristics of the reservoir. This enables oil and gas companies to identify potential production problems and optimize well performance.

Enhanced Oil Recovery (EOR): EOR techniques involve injecting fluids such as water, gas, or chemicals into the reservoir to improve the recovery of hydrocarbons. EOR techniques can significantly increase the production from mature reservoirs and extend the life of the reservoir.

Intelligent Well Systems: Intelligent well systems use sensors and communication technology to monitor and control the production from individual wells in real-time. This enables oil and gas companies to optimize the performance of each well and reduce the overall cost of production.

Digital Oilfield: The digital oilfield involves using advanced technologies such as data analytics, machine learning, and artificial intelligence to optimize the performance of oil and gas operations. The digital oilfield enables companies to monitor and control the performance of reservoirs, wells, and production facilities in real-time, which can significantly improve efficiency and productivity.

2.2 Well Construction

Well construction is a critical aspect of the oil and gas industry, as it involves the design, drilling, and completion of wells that are used to extract hydrocarbons from the reservoir. Well construction is a complex process that requires advanced technologies and frameworks to ensure the safe and efficient extraction of hydrocarbons.

Some key technologies and frameworks used in well construction in the oil and gas industry include:

Drilling Technology: Drilling technology has undergone significant advancements over the years, and today's drilling rigs are equipped with advanced sensors and control systems that enable precise and efficient drilling. Technologies such as rotary steerable systems, logging-while-drilling, and measurement-while-drilling enable real-time monitoring of the drilling process, which can improve efficiency and reduce the risk of drilling problems.

Wellbore Design: Wellbore design involves determining the optimum size and shape of the wellbore, which can affect the efficiency and productivity of the well. Advanced wellbore

design techniques such as hydraulic fracturing and horizontal drilling can significantly increase the production of hydrocarbons from the well.

Completion Technology: Completion technology involves the process of preparing the well for production, which includes the installation of casing, tubing, and other equipment. Advanced completion technologies such as multi-zone completion and intelligent well completion can optimize the production of hydrocarbons from the well and improve efficiency.

Well Integrity Management: Well integrity management is a critical aspect of well construction that involves ensuring the safety and reliability of the well throughout its life cycle. Advanced technologies such as acoustic and electromagnetic surveys, fiber-optic sensors, and real-time monitoring systems can detect and prevent well integrity issues, which can improve safety and reduce downtime.

Environmental Protection: Well construction must also consider environmental protection, and technologies such as closed-loop drilling systems and recycling of drilling fluids can significantly reduce the environmental impact of drilling operations.

2.3 Converters

Some key types of converters used in the oil and gas industry include:

Variable Frequency Drives (VFDs): VFDs are used to control the speed and torque of electric motors used in pumps, compressors, and other equipment. VFDs can significantly improve energy efficiency and reduce maintenance costs.

Uninterruptible Power Supplies (UPS): UPS systems are essential in the oil and gas industry, where downtime can have significant financial and safety consequences.

Power Inverters Power inverters are used in remote oil and gas sites that are not connected to the grid.

Static VAR Compensators (SVCs): SVCs are used to stabilize the voltage and improve the power factor of electrical systems. SVCs are essential in the oil and gas industry, where voltage fluctuations can cause equipment damage and downtime.

Active Harmonic Filters: Active harmonic filters are used to eliminate harmonic distortion in electrical systems. Harmonic distortion can cause equipment damage and reduce the efficiency of electrical systems.

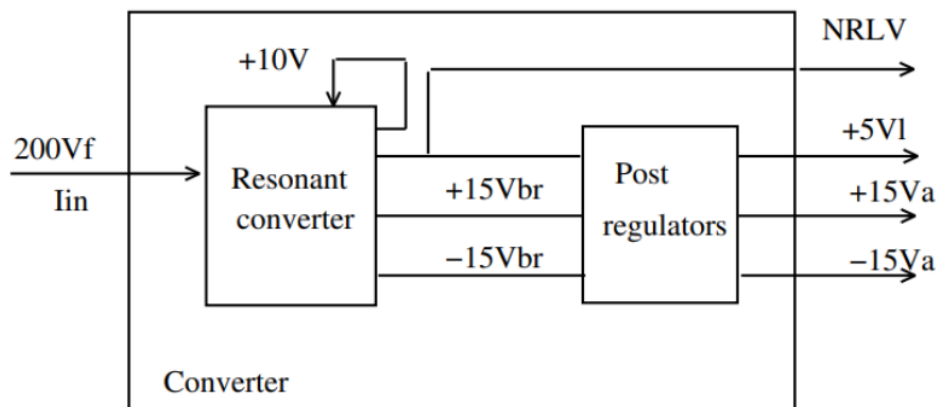
CHAPTER 3

PROJECT ARCHITECTURE AND DESIGN

3.1 PROJECT

CONVERTERS

The conversion is achieved through high frequency switching using inductive and capacitive filter elements. Power converters are designed to optimize the electric energy for a specific load by changing it from one form to another. Depending on the application, these functions can include increasing or decreasing the input voltage magnitude, inverting polarity, or producing multiple output voltages of the same or different polarity. For example, computer power supply units use DC-DC converters to generate mixed polarity output voltages.



- $200V_f$: Input voltage.
- I_{in} : input current
- +10V: Internal converter voltage
- +15Vbr: Internal converter voltage.
- -15Vbr: Internal converter voltage.
- +15Va: Output voltage.

- -15Va: Output voltage.
- NRLV: Output converter voltage.
- +5Vl: Output converter voltage.
- fs: converter switching frequency.

A DC-DC converter is an electrical circuit that converts a direct current (DC) voltage from one level to another. The main purpose of a DC-DC converter is to take an input DC voltage and convert it to a different, typically lower or higher, DC voltage that is suitable for a specific application. The converter operates on the principle of energy transfer, whereby electrical energy is transferred from the input to the output of the circuit with a minimum of power loss.

There are many different types of DC-DC converters, each with their own advantages and disadvantages. Some common types of DC-DC converters include buck converters, boost converters, buck-boost converters, and flyback converters.

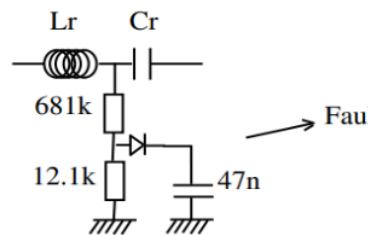
Buck converters are used to step down the input voltage to a lower output voltage. They are typically used in applications where high efficiency is required and the output current is less than the input current. Boost converters, on the other hand, are used to step up the input voltage to a higher output voltage. They are typically used in applications where a higher voltage is required than the input voltage, such as in battery-powered devices.

Buck-boost converters are used to step up or step down the input voltage to a different output voltage. They are typically used in applications where the input voltage can vary widely and a stable output voltage is required. Flyback converters are used to provide electrical isolation between the input and output of the circuit, making them useful in applications where high voltage isolation is required, such as in power supplies for medical equipment.

DC-DC converters are used in a wide range of applications, including power supplies for electronic devices, battery chargers, renewable energy systems, and motor control systems. They are available in a variety of power levels, from low-power converters for microelectronics to high-voltage converters for power transmission systems

short circuit protection

The short circuit protection is made by the detection of an over voltage at the node between the resonant inductance and the resonant capacitance. This stops the operation of the converter that initiates a start-up sequence. At power up, the charge of the load filtering capacitors produces a current peak, and the detection of an over voltage must be delayed. This is obtained by using high resistor values and capacitive low pass filtering.



short circuit protection

COMPONENTS USED

(1) Power Capacitor

A power capacitor is an electrical component that can store and release electrical energy. It is made up of one or more pairs of plates separated by an insulating material called a dielectric. These plates are connected to two terminals that enable the stored energy to be discharged into an electrical circuit when needed. Power capacitors are commonly used in industrial applications to improve the power factor of electrical systems that have inductive loads, such as those generated by induction motors. Poor power factor can result in inefficient energy consumption and negative effects on distribution lines.

Diodes

(1) Low-power voltage regulator diodes

Low-power voltage regulator diodes in small hermetically sealed glass SOD80C SMD packages.

(2) Switching diode

Is a high-speed switching diode fabricated in planar technology.

(3) Ultra-fast low-loss controlled avalanche rectifiers.

Cavity free cylindrical glass SOD87 package through Implotec technology

MOSFETS

(1) N-Channel JFET Switching Transistors.

The junction-gate field-effect transistor (JFET) is one of the simplest types of field-effect transistor.

(2) Power MOSFET

Third generation power MOSFETs from Vishay provide the designer with the best combination of low on-resistance, and cost-effectiveness.

FEATURES

- Dynamic dv/dt rating
- Repetitive avalanche rated

(3) Power MOSFET

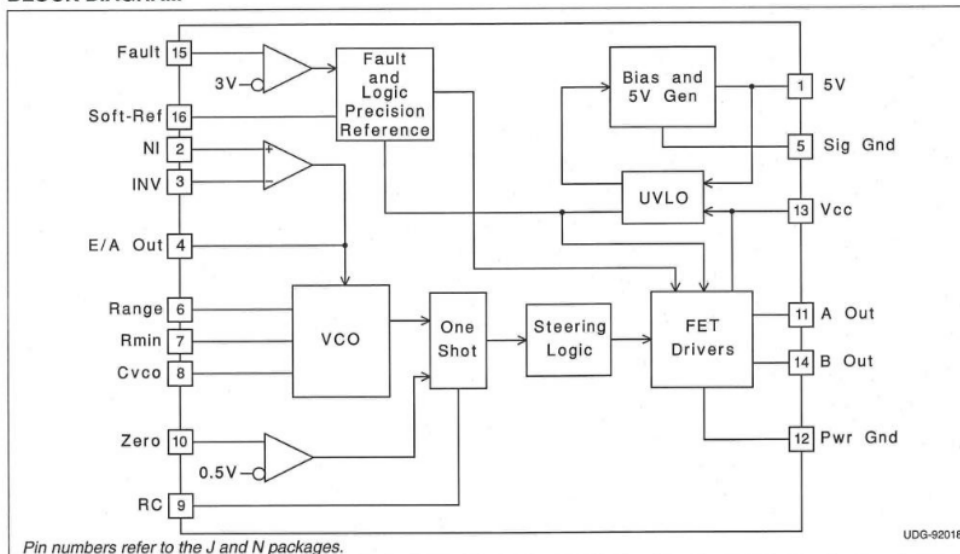
N-channel power MOSFET that can switch loads up to 400V and consume up to 10A with a gate threshold voltage of 10V across the gate and source pins could be suitable for many power electronics applications. However, it is important to select a specific MOSFET device that

meets the requirements of your application and to properly design the circuit to ensure reliable and efficient operation.

(1) Resonant-Mode Power Supply Controller

This group of ICs is designed to control quasi-resonant converters that utilize Zero Current Switching or Zero Voltage Switching. The different variations within the family are determined by the the VCO. This allows for "true" zero current or voltage switching, while also accommodating the initial tolerances of the resonant components and various line, load, and temperature changes. Under-Voltage Lockout ensures safe starts upon power-up, with a supply current during this period typically below 150mA and actively forced low outputs.

BLOCK DIAGRAM



BLOCK DIAGRAM

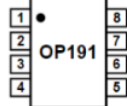
The has UVLO thresholds of 16.5V (ON) and 10.5V (OFF), while the thresholds for the are 8V (ON) and 7V (OFF). Once Vcc exceeds the UVLO threshold, a 5V generator is enabled to provide bias for internal circuits and up to 10mA for external use

(2) Amplifiers

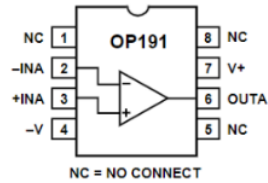
Applications

- Industrial Process Control.
- Battery Powered Instrumentation.

**8-Lead Narrow-Body SO
(S Suffix)**



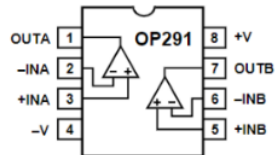
**8-Lead Epoxy DIP
(P Suffix)**



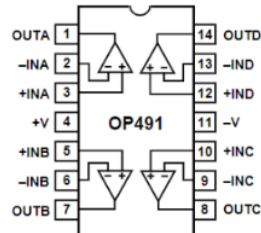
**8-Lead Narrow-Body SO
(S Suffix)**



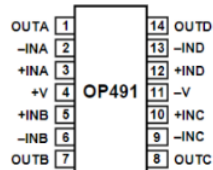
**8-Lead Epoxy DIP
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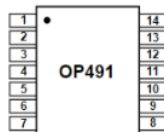
**14-Lead Epoxy DIP
(P Suffix)**



**14-Lead SO
(S Suffix)**



**14-Lead
TSSOP
(RU Suffix)**



PIN DESCRIPTION

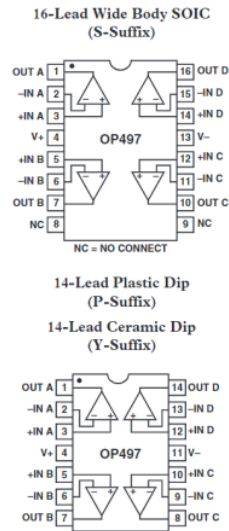
Description

The micropower amplifiers family includes single, dual, and quad versions that feature a 3 MHz bandwidth and rail-to-rail inputs and outputs. The output voltage swings within millivolts of the supplies, allowing it to sink or source current all the way to the supplies. An operational amplifier, or op amp, is a type of electronic circuit that is commonly used in analog circuits for signal amplification and filtering. It is a high-gain, differential amplifier with a very high input impedance and a low output impedance. Op amps are typically constructed from transistors and other electronic components, and they are available in a variety of configurations and package types.

Op amps have two input terminals, one labeled as the inverting input (-) and the other labeled as the non-inverting input (+). The output of the op amp is typically taken from a third terminal labeled as the output. The output of the op amp is proportional to the difference between the voltages at the two input terminals. In other words, the output of the op amp is equal to the gain of the amplifier multiplied by the voltage difference between the two input terminals.

Op amps are typically used in a wide range of analog circuits, including amplifiers, filters, oscillators, and signal generators. They are also used in digital circuits, such as comparators and voltage regulators. Op amps are often used in combination with other electronic components, such as capacitors, resistors, and diodes, to create more complex circuits with specific functions and characteristics.

(3) Amplifiers

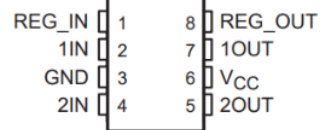


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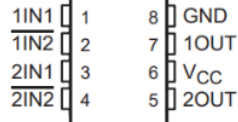
Description

The quad op amp is designed to deliver precise performance while occupying minimal space, thanks to its industry-standard 16-lead SOIC package, with a supply current of less than 625 μA .

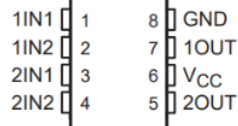
**TPS2811, TPS2812, TPS2813 . . . D, P, AND PW
PACKAGES
(TOP VIEW)**



**TPS2814 . . . D, P, AND PW PACKAGES
(TOP VIEW)**



**TPS2815 . . . D, P, AND PW PACKAGES
(TOP VIEW)**

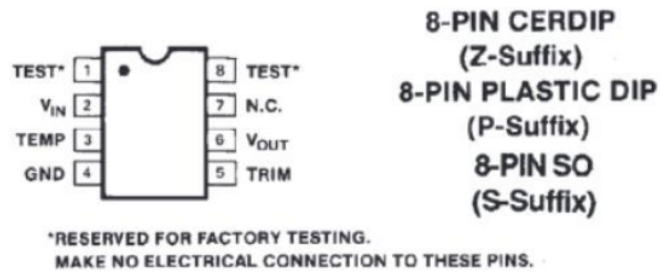


PIN DESCRIPTION

Description

The dual high-speed MOSFET drivers series can manage highly capacitive loads and supply peak currents of up to 2 A. This superior performance is attained due to a design that inherently reduces shoot-through current and consumes significantly less supply current compared to competing products. These drivers include a regulator that allows them to operate with supply inputs ranging from 14 V to 40 V. The output of the regulator can also be used to power other circuitry, as long as the They are equipped with 2-input gates, providing greater flexibility in MOSFET control

(4) low power precision reference



PIN DESCRIPTION

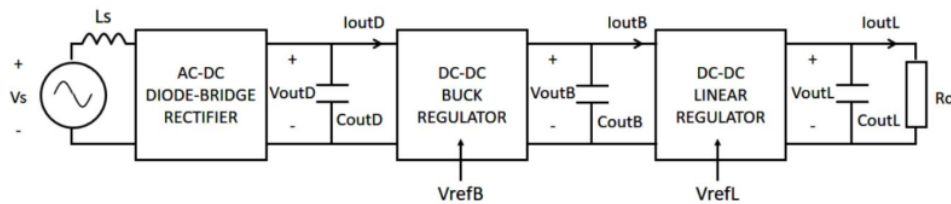
Description

A precision reference that operates at low power is capable of providing a constant output of 2.5 V, regardless of changes in load conditions, ambient temperature, or supply voltage. This reference level is applicable, as well as any other applications that necessitate a reliable and consistent voltage. The output voltage and its temperature coefficient are zener-zap trimmed to guarantee precise output tolerances and low thermal drift. Additionally, a curvature correction circuit is utilized to decrease the thermal curvature that was commonly observed in previous bandgap references.

CONVERTER

General information,

- 200Vf: Input voltage. Nominal range value: 165V, 225V
- I_{in} : input current
- +10V: Internal converter voltage.
- +15Vbr: Internal converter voltage.
- -15Vbr: Internal converter voltage.
- +15Va: Output voltage.
- -15Va: Output voltage.
- +5Vl: Output converter voltage.
- f_s : converter switching frequency.



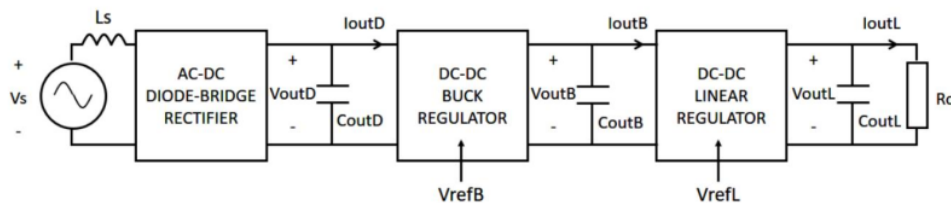
ANALYSIS

1. 200Vf: Input voltage. Nominal range value: 165V, 225V.
2. DC to DC converter.
3. RTN is return.
4. +10V is used as feedback in the circuit.
5. Resistors acts as a voltage divider.
6. Net value of all the capacitance will be added in a parallel manner.
7. There is a power capacitor to charge the MOSFET.
8. There is Voltage regulator diodes. Protects the transistor, reverse breakdown voltage.
9. The point between inductance and capacitance is the point for short circuit detection.

10. Ultra-fast low-loss controlled avalanche rectifier which act as a center tapped full wave rectifier, uses a centre tapped transformer and two diodes to convert the complete AC to DC signal).
11. Ultra-fast low-loss controlled avalanche rectifier act as a full bridge with 4 diodes to convert AC to DC and to reverse the direction of AC pulse.
12. High speed diode to cut off the signal.
13. Resistors have half equivalent voltage.
14. There is a filter capacitor where the input voltage is gathered as in the form of sinusoidal and when it is discharging the output signal will be in the form of ripple wave.
15. Capacitors are in parallel to it. It is given to avoid sinusoidal wave in DC where the expected signal wave will be linear.
16. Voltage regulator diodes, diode potential voltage drop.
17. There is High speed diode to not interfere in fault line.
18. VCC is getting +10V near the test point, along with Voltage regulator diodes, diode potential voltage drops and it will pass AC.
19. Reg-In is connected to Reg-Out.
20. Two inputs are connected from A_OUT and B_OUT of U6-UC1863.
21. In IC TPS2812, VCC, SIG_GND and PWR_GND are both connected to ground.
22. SOFT_REF, INV, E/A_OUT relate to resistors along with capacitor.
23. SV thus completes RC with the resistor thus connected to the ground with.
24. The FAULT is connected with as the voltage divider (from the input +200VF) followed by high-speed diode to not interfere in fault line along with acting as DC block and AC bypass.
25. Short circuit protection the short circuit protection is made by the detection of an over voltage at the node between the inductance and the capacitance. This stops the operation of the converter that initiates a start-up sequence. At power up, the charge of the load filtering capacitors produces a current peak, and the detection of an over voltage must be delayed. This is obtained by using high resistor values and capacitive low pass filtering.
26. Output converter voltage.
27. The output for the resonant converter will be,
 - +10V: Internal converter voltage. Nominal value: 11.8 V
 - +15Vbr: Internal converter voltage. Nominal value: 17.4V
 - -15Vbr: Internal converter voltage. Nominal value: -17.4V

REGULATOR

- $200V_f$: Input voltage.
- I_{in} : input current
- $+10V$: Internal converter voltage.
- $+15V_{br}$: Internal converter voltage.
- $-15V_{br}$: Internal converter voltage
- $+15V_a$: Output voltage.
- $-15V_a$: Output voltage.
- $+5V_l$: Output converter voltage.
- f_s : converter switching frequency.



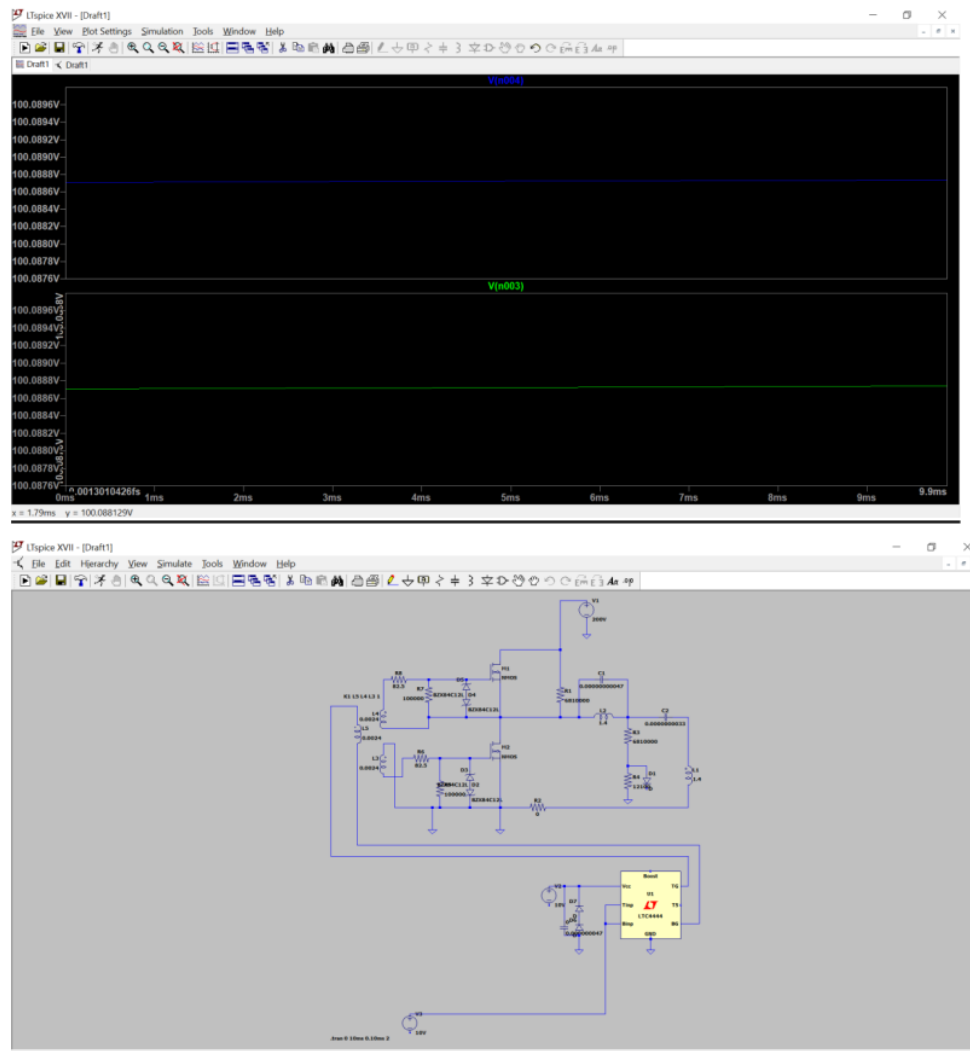
BLOCK DIAGRAM

ANALYSIS

1. Three regulators are present in a row with equivalent resonance input and output.
2. The resonant circuit is to provide the input filter in which group of capacitors are used to filter the input voltage and remove any high-frequency noise.
3. The reference voltage is achieved from the IC which has input from the resonant converter, which is set to be as $+15V_{br}$, $-15V_{br}$ and $NRLV$.
4. There is low power precision reference which provides reference voltage input to the error amplifier or op amp.
5. There is used as a reference voltage to the top left followed by the next regulator right.
6. There from is used as a reference voltage inverting input to the top right followed by the next regulator left.

7. The transistors are JFET, junction-gate field-effect transistor is used to build the amplifier.
8. The regulators are built with common source ballast MOS transistors, to allow linear operation with dropout voltages.
9. The common source ballast transistor presents a voltage at its drain equal to the transconductance times the load impedance times the gate to source voltage.
10. The ballast MOS transistor includes as third generation power MOSFET.
11. There is a N-channel power MOSFET.
12. The op-amps used as error amplifiers will include a dominant pole compensation.
13. The OP497 is a quad op amp with precision performance in the space-saving.
14. Dual micropower, single-supply, 3 MHz bandwidth amplifiers featuring rail-to-rail inputs and outputs.
15. The inputs of the current limiting op-amp are biased near the nominal regulator output voltage. When in current limiting, they are biased near zero volts.
16. The output of the amplifier is normally near ground and rises to 2.5 volts plus the forward diode bias when current limiting.
17. Each of the three regulators has an independent current limiter to protect the load and to allow easier debugging of faults.
18. These limiters exhibit a fold back characteristic to reduce the power dissipated by the ballast transistor.
19. Parasite ripple is generated by the 15Va fold-back amplifier at a frequency from 300Hz to 3kHz and has a 400mVpp of amplitude max comparison of the parasite ripple with the fold-back current detection and without it.
20. A Zener diode on the feed-back avoids saturation and prevent this ripple. This has not been implemented because there is no practical impact in the performance of the regulator.
21. The output ripple frequency is the result of the full wave rectifying of the alternating resonant current of the resonance converter.

LT Spice Simulation



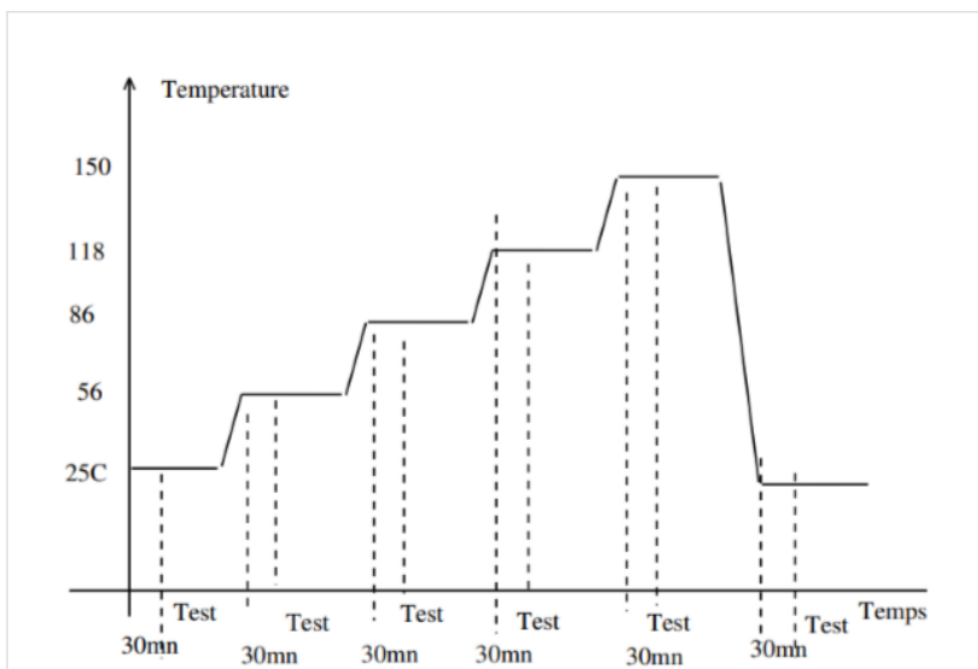
Here we can see is the simulation of the particular part of the circuit which I have run on LT spice with three major graphs as shown with both the transistor and,

- TRANSFORMER 1- RESISTANCE: $R_{1-2} = R_{3-4} = 0,55^\circ$ $R_{5-6} = 0,55^\circ$ $R_{7-8} = 0,550$
- INDUCTANE- 1.4H, SERIES RESISITANCE- 1.4O

- TRANSFORMER 2- RESISTANCE: R 1-2 = 1,4° R 4-5 =R 6-7= 45mO, R 9-10 =R 11-12= 0,22° , R 14-15 =R 15-16= 0,5°
- INDUCTANCE- 0.0024H, SERIES RESISTANCE- 0.55O

TESTS

1. Test conditions: The board must be supplied between Test point following the 3 test conditions successively: 165Vdc, 200Vdc and 225Vdc.
The tests must be executed at room temperature and 150C.
1. These tests will be carried out on a board partially equipped without these transistors must be assembled on the heater beforehand.
2. All accessible equipotential nodes will be tested the ones with others to eliminate short-circuits induced eventually by solder jumpers or metal particles but also the continuity of connections between components.
3. All passive components which can isolated electrically from the others, and which do not require the power supply application will be tested within the limit of their tolerances. The parallel components which cannot be insulated will be tested like only one component.
4. The digital components (UC1863, TPS2812) having a model into the standard library of tester will be tested by power supply & stimulus applying on input pins and control of conformity on output pins.
5. From the standard library of tester, it will be developed specific models for the analog components (REF43, OP291, OP497) used on this board to check their good performance and functionality.
6. Mean view of power supply board with its heater: The high voltage connection (up to 225Vdc) must ensure good removing contacts but also an electric protection in order to avoid all contacts of the operator during the tests. The integrity of this protection must be guaranteed up to 175C.
7. Short circuit current measurement, outputs short circuit measurement are made with a variable load on each output. +15Va: Results at 25C: At 150C the measured short-circuit current is 24mA.



profile characterization

CHAPTER 5

LEARNING SKILLS AND EXPERIENCES

5.1 Technical Enhancement

As part of my technical enhancement in the field of failure analysis of PCB and PWA, I have gained the following skills and knowledge:

PCB and PWA Design: I have learned about the different types of circuits, components, and materials used in PCB and PWA design, as well as how to read schematics and interpret design specifications. This has helped me to better understand the potential causes of failures and to identify any design issues that may be contributing to failures.

Failure Modes and Analysis Techniques: I have developed technical expertise in the different failure modes that can occur in PCBs and PWAs, such as solder joint failures, component failures, and board-level failures. I have also learned about various analysis techniques, such as visual inspection, X-ray inspection, and microsection analysis, that can be used to identify the root cause of failures.

Test and Measurement Techniques: I have gained knowledge of various test and measurement techniques that can be used to evaluate the performance of PCBs and PWAs, such as voltage and current measurements, impedance measurements, and signal integrity measurements. This has helped me to better understand the performance of PCBs and PWAs and to identify any issues that may be impacting their performance.

Data Analysis: I have developed skills in data analysis, which is important in order to identify trends and patterns in failure data. This has allowed me to identify any recurring issues or patterns that may be contributing to failures and to develop strategies to address these issues.

Overall, my technical enhancement in the field of failure analysis of PCB and PWA has allowed me to better understand the causes of failures, to identify potential issues, and to develop

effective strategies to address these issues. I am confident that these skills will be valuable in my future work as an electronic engineer.

5.2 Business Terminologies

As part of my learning in the field of failure analysis of PCB and PWA, I have also gained knowledge of business terminologies that are commonly used in the industry. Some of the key business terminologies that I have learned include:

Root Cause Analysis: This is a systematic approach used to identify the underlying causes of a problem or failure. By identifying the root cause, it is possible to develop more effective solutions that can prevent the problem from recurring.

Cost of Quality: This refers to the costs associated with ensuring that products or services meet the desired level of quality. These costs can include the cost of inspections, testing, and rework.

Continuous improvisation: Is an ongoing effort to improve processes and products in order to increase efficiency, reduce costs, and improve quality. In the context of failure analysis of PCB and PWA, continuous improvement is essential to ensure that processes are optimized and that any issues are addressed in a timely and effective manner.

14 Supply Chain Management: This refers to the processes used to manage the flow of goods and services from suppliers to customers. In the context of failure analysis of PCB and PWA, effective supply chain management is essential to ensure that the right components and materials are sourced from reliable suppliers and that products are delivered to customers on time and at the desired level of quality.

Overall, my knowledge of these business terminologies has helped me to better understand the broader context in which failure analysis of PCB and PWA takes place, and to appreciate the importance of effective processes and quality control in ensuring that products meet the desired level of quality.

5.3 Business Essentials

In addition to technical skills, my learning in the field of failure analysis of PCB and PWA has also included important business essentials. Some of the key business essentials that I have learned include:

Communication Skills: Effective communication is essential in any business setting, and particularly in the context of failure analysis of PCB and PWA. It is important to be able to communicate issues, solutions, and progress clearly and concisely to stakeholders at all levels of the organization.

Project Management: Failure analysis of PCB and PWA often involves complex projects with many stakeholders and moving parts. Understanding the principles of project management, including planning, resource allocation, and risk management, is essential to ensure that projects are completed on time and within budget.

Collaboration: Failure analysis of PCB and PWA requires collaboration between different teams and departments within an organization, as well as with external suppliers and customers. Understanding how to work effectively in a team and how to build strong relationships with stakeholders is essential for success.

Business Ethics: Conducting failure analysis of PCB and PWA requires a high degree of ethical behavior, including respect for intellectual property, compliance with relevant regulations and standards, and transparency in reporting. Understanding the principles of business ethics is essential to ensure that work is conducted in a responsible and professional manner.

Financial Management: Understanding the financial implications of failure analysis of PCB and PWA, including the costs associated with testing and rework, is essential for effective decision making. This includes understanding financial statements, budgeting, and cost-benefit analysis.

Overall, my understanding of these business essentials has helped me to appreciate the broader context in which failure analysis of PCB and PWA takes place, and to understand the

importance of effective communication, collaboration, and ethical behavior in ensuring success.

CHAPTER 6

CONCLUSION

In conclusion, my role as a failure analysis engineer in the field of PCB and PWA at Schlumberger has been instrumental in my professional growth and development. Through my work in analysing failures and implementing solutions, I have gained valuable technical skills and knowledge in areas such as PCB layout design, electronic components, and testing methodologies. Additionally, my experience has allowed me to develop important business essentials such as communication skills, project management, collaboration, business ethics, and financial management.

My internship as a Failure Analysis Engineer at Schlumberger provided me with a valuable learning experience. I had the opportunity to work on troubleshooting a board related to the reservoir performance product line, where I gained hands-on experience in performing failure analysis and utilizing various tools, including schematic analysis and software like Itspice.

During my time at Schlumberger, I learned a lot about the steps involved in troubleshooting any PCB or PWA, including identifying the issue, developing a hypothesis, testing, and validating the solution. This knowledge will undoubtedly prove useful in my future career as a failure analysis engineer.

Although there were some challenges along the way, the internship allowed me to develop my technical skills and problem-solving abilities. I also learned the importance of effective communication and collaboration in a team environment, which is critical in the field of engineering.

Overall, I am grateful for the opportunity to have worked at Schlumberger and gained practical experience in failure analysis. I am confident that the skills and knowledge I acquired during my internship will serve me well in my future endeavours as an engineer.

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